UNIT 2 · TOPIC 1(B)

Belt Drives and Pulleys

Syllabus topic: Design and selection of machinery elements: gears, pulleys, chains and sprockets, belts

Flat belts

- Flat belts transmit power by friction. Crown the pulley face to keep the belt tracking centrally.

$$T_1/T_2 = e^{(k\theta)} \text{ (Euler/Eytelwein; } \theta = \text{arc of contact in rad, } k = \text{coefficient of friction)}$$

- For rubber flat belts on steel/CI pulleys, **k ≈ 0.25–0.30**, giving a tension ratio of **≈ 2.2–2.6** at 180° arc (Kepner §4.3).

$$P = (T_1 - T_2) \times v$$

$$hp = S(T_1 - T_2)/33{,}000 \text{ (Kepner eq. 4.1, fpm/lb units)}$$

- Centrifugal tension **T_c = m·v²** (m = belt mass per unit length) adds to both sides and reduces useful capacity.
- Power is maximum when **$T_1 = 3T_c$**, giving the optimum speed **$v = \sqrt{(T_1/3m)}$**.
- **Slip** = relative motion between belt and pulley (2–3% typical). **Creep** = elastic shortening/lengthening as the belt passes over the pulleys — it happens even without overload.

V-belts (Kepner §4.9–4.13; ASAE standards)

- Wedging in the sheave groove multiplies effective friction, so V-belts work with small arcs of contact and short centre distances — a key advantage on serpentine combine drives.
- Belt cross-section included angle = **40°**; sheave groove angle **34–38°**. The groove is narrower than the belt so the belt wedges; smaller sheaves get smaller groove angles to compensate for belt distortion.
- Design tension ratio for V-belts = **5** (k ≈ 0.51, including wedging) — Kepner §4.3. Larger ratios risk instability from low slack-side tension.
- A V-belt sheave and a flat pulley are "balanced" in capacity when the arc of contact is **≈ 130°** on the sheave and **230°** on the flat pulley (equal kθ) — the basis of V-flat drives.
- **Arc correction factor:** ratings assume a 180° arc; reduce capacity for smaller arcs (Kepner §4.4). A **service factor** accounts for load character (smooth → high shock) and expected life.

$$L \approx 2C + (\pi/2)(D + d) + (D - d)^2/(4C) \text{ (open drive)}$$

Solve the standard quadratic of this equation for the centre distance C.

Belt sections	Designations
Industrial (classical)	A, B, C, D, E
Agricultural (ASAE)	HA, HB, HC, HD, HE — built for higher flexing life and dirt/moisture conditions
Double (hexagonal)	HAA, HBB, HCC — serpentine drives with reverse bends

- **Variable-speed V-belt drives:** wide belts on adjustable-pitch sheaves change speed on the go (combine ground drive); the pitch-diameter range typically gives **2–4×** speed variation (Kepner §4.12).
- **Banded (joined) belts** prevent whip and turnover in multi-belt drives. Match (length-grade) the belts in multiple-belt drives.

Belt tensioning

- Use idlers (inside/outside, tight/slack side), spring-loaded idlers as belt clutches (to engage/disengage the drive) and automatic take-up.
- Place an idler on the slack side near the small pulley to increase the arc of contact.

Practice MCQs

Q2. In a belt drive, the ratio T_1/T_2 equals:

(a) e^kθ (b) kθ
(c) e^θ/k (d) log(kθ)

Q3. The tension ratio commonly assumed in designing V-belt drives is:

(a) 2 (b) 3
(c) 5 (d) 8

Q4. For rubber flat belts on steel pulleys, the tension ratio at 180° arc of contact is about:

(a) 1.2–1.5 (b) 2.2–2.6
(c) 4–5 (d) 6–8

Q5. The included angle of a standard V-belt cross-section is:

(a) 30° (b) 34°
(c) 38° (d) 40°

Q6. Sheave groove angles are kept at 34–38°, i.e., less than the belt angle, to:

(a) Reduce belt speed (b) Ensure wedging action
(c) Increase slip (d) Ease belt removal

Q7. A V-belt sheave and a flat pulley are balanced in power capacity when arcs of contact are about:

(a) 180° and 180° (b) 130° and 230°
(c) 230° and 130° (d) 90° and 270°

Q8. Belt power is maximum when centrifugal tension equals:

(a) T_1 (b) $T_1/2$
(c) $T_1/3$ (d) $3T_1$

Q9. Creep in belts is caused by:

(a) Overload slip (b) Alternate elastic elongation/contraction of the belt over pulleys
(c) Misalignment (d) Wear

Q10. Agricultural V-belt cross-sections per ASAE standards are designated:

(a) A, B, C (b) HA, HB, HC, HD, HE
(c) SPA, SPB (d) 3V, 5V, 8V

Q11. Double (hexagonal) V-belts are used for:

(a) Higher speeds (b) Serpentine drives with reverse bends
(c) Vertical shafts (d) Wet conditions

Q12. Flat-belt pulleys are crowned to:

(a) Increase friction
(b) Keep the belt tracking at pulley centre
(c) Reduce weight
(d) Balance the pulley

Q13. An idler pulley placed on the slack side near the small pulley:

(a) Reduces arc of contact
(b) Increases arc of contact and capacity
(c) Increases slip
(d) Reduces belt life only

Q14. Variable-speed V-belt drives change speed by:

(a) Changing motor speed
(b) Adjustable-pitch sheaves altering effective pitch diameter
(c) Gearbox
(d) Slipping clutch

Answers

Q2 — (a) e^kθ

Q3 — (c) 5 — *Kepner §4.3 — flat rubber belt k = 0.25–0.30 gives R = 2.2–2.6 at 180°; V-belt design assumes R = 5 (k ≈ 0.51 including wedging).*

Q4 — (b) 2.2–2.6 — *Kepner §4.3 — flat rubber belt k = 0.25–0.30 gives R = 2.2–2.6 at 180°.*

Q5 — (d) 40°

Q6 — (b) Ensure wedging action

Q7 — (b) 130° and 230° — *Equal kθ values → 130° sheave ≈ 230° flat pulley (Kepner §4.9); basis of V-flat drives on threshers.*

Q8 — (c) $T_1/3$ — *dP/dv = 0 with P = (T_1 -T_c-T_2')v … yields T_c = T_1/3.*

Q9 — (b) Alternate elastic elongation/contraction of the belt over pulleys

Q10 — (b) HA, HB, HC, HD, HE

Q11 — (b) Serpentine drives with reverse bends

Q12 — (b) Keep the belt tracking at pulley centre

Q13 — (b) Increases arc of contact and capacity

Q14 — (b) Adjustable-pitch sheaves altering effective pitch diameter

Sources: Kepner, Bainer & Barger — Principles of Farm Machinery, Ch. 4 (Elements of Rotary Power-Transmission Systems) and Appendix (ASAE V-belt standards); standard machine design references (Shigley; V.B. Bhandari); ASAE/IS standards.